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Robotic System for Monitoring Industrial Installations



Figure 1: AI generated depiction of the finished project

Progress Report of the Final Year Project

Bachelor's Degree in Electronics, Telecommunications and Computer Engineering
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Chapter 1

Introduction

This chapter presents challenges of integrating a thermal imaging system into a quadruped robotic platform for industrial monitoring applications. It defines the problem addressed, establishes the project objectives and scope, and introduces the key technologies involved, including thermal sensing and robotic platforms.

1.1 Overview

This project aims to use a mobile robotic platform (*Unitree Go2* quadruped model) and integrate a thermal camera, or another type of sensor that may be relevant during the project for use in monitoring industrial installations. This will allow, in a safe and desirably autonomous way, for the performance of inspections of critical equipment, such as electrical substations. With this system, the aim is to develop the ability to control the robot, receive and monitor images transmitted via the Internet using a wireless network. In the end, the project aims to demonstrate a use case of industrial inspection with an application close to a real-world scenario.

1.2 Motivation

The motivation for this project lies in improving the safety and efficiency of industrial inspections. Thermal imaging is widely used to detect early signs of equipment failure, but current inspection methods are manual and expose operators to hazardous conditions. By integrating a thermal camera into a quadruped robotic platform, it becomes possible to perform remote and repeatable inspections, reduce human risk, and enable more frequent monitoring of critical infrastructure.
As *Unitree*'s *Go2* robot is very cheap (from around 1.300€ to 10.000€ depending on the version[1]) in comparison with other robots in the market, like *Boston Dynamics*' *spot* at around 60.000€[2] and *Anybotics' Anymal* at 130.000€[3] with the both having a option for a thermal camera integration (**Figure 2** and **Figure 3**), having a cheap alternative to these robots would allow thermal inspection solutions at a significantly lower cost, making them accessible beyond large industrial operators. This opens the possibility of deploying multiple units for increased coverage, performing more frequent inspections, and experimenting with custom sensing and control architectures without the financial constraints associated with high-end platforms.



Figure 2: Anybotics' Anymal with a thermal camera add-on



Figure 3: Boston Dynamics' Spot with a thermal camera add-on

1.3 Project Architecture

For the creation of this project it has to be taken into account not only what components to integrate but how they communicate with each other.

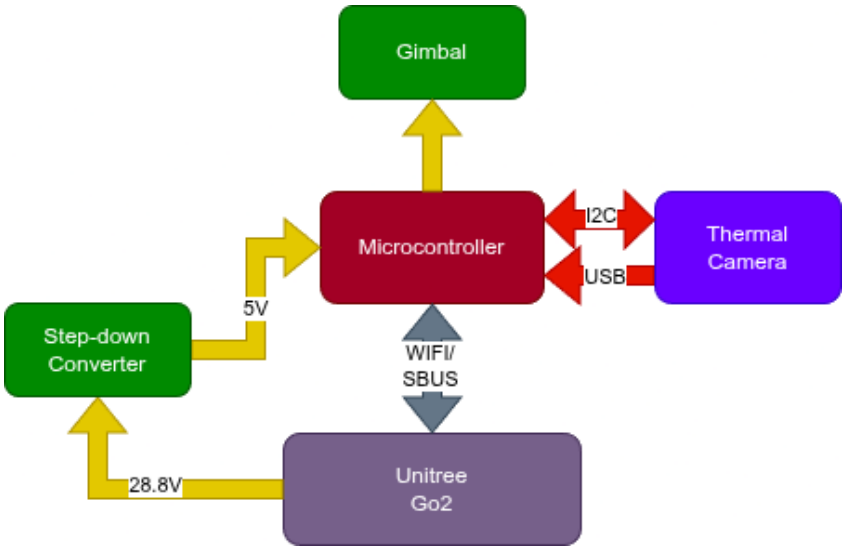


Figure 4: Block Diagram of Project Architecture

1.3.1 Components

- Unitree Go2 (quadruped)
- Thermal camera
- Gimbal
- Micro-controller
- DC-DC step-down converter (28.8V to 5V/3.3V)

1.3.2 Details

The Unitree Go2 robot provides 3 ports: RJ45, XT30, S-BUS.[4]

- RJ45 is a port for ethernet networking, and may or may not be used, as the robot has the ability to be controlled via WIFI using the webRTC API [5].
- The XT30 connector provides 28.8V directly from the Go2 battery, but may vary between 24V to 33.6V depending on the state of charge.[4] This has to be taken into account when choosing a step-down module to convert to a lower voltage.
- S-BUS is a port that only receives remote control data and can only can handle the commands that are available on the handheld remote control provided by Unitree [6]. It may or may not be needed as the webRTC API provides the same functionality.

1.3.3 Requirements

The thermal camera should preferably be radiometric (can measure absolute temperature) and LWIR (Long Wave Infrared) as these cameras can detect sufficiently high temperatures for the scope of this project, and have the benefit of not needing active cooling, unlike MWIR (Medium Wave Infrared). [7]

To mount the thermal camera onto the robot it would preferable to have a gimbal as it would allow precise aiming without the need of moving the robot.

The most suitable micro-controller for this project would be one with I2C, USB, WIFI/Ethernet and enough bandwidth to send video with a resolution of 640x512 at 50Hz.

The DC-DC step-down converter must provide a stable 5V output for the camera and micro-controller while taking into account the variability of the XT30 port as referenced in **Section 1.3.2**

1.4 Problems to solve

1.4.1 Finding a Thermal Camera

Maybe the most important part of the project is finding and choosing a thermal camera, which presents a significant challenge, as most cameras need inquiries to find the price, and are very expensive (in the thousands of Euros).

Type	Name	Radio-metric	Temp. range (High Gain)	Temp. range (Low Gain)	Video Data	Control Prot.	Supply Voltage	Power dissipation (operating/ standby)	FOV Diag.	FOV Horiz.	Resolution (HxV)	Frame rate (Hz)	Price
module	Lepton UWFOV	no	−10°C to 140°C ±5°C	−10°C to 450°C ±10°C	SPI	CCI (I2C)	2.8V - 3.1V	140mW / 5mW	160°	160°	160×120	9	120€
module	Lepton 3.5	yes	−10°C to 140°C ±5°C	−10°C to 450°C ±10°C	SPI	CCI (I2C)	2.8V - 3.1V	140mW / 5mW	71°	57°	160×120	9	138€
module	Lepton 3.1R	yes	−10°C to 140°C ±5°C	−10°C to 450°C ±10°C	SPI	CCI (I2C)	2.8V - 3.1V	140mW / 5mW	119°	95°	160×120	9	142€
camera	S302NP	no	−40°C to +330°C	-	USB	-	3.3 - 5.0V	300mW	-	105°	320×240	<9	421€
camera	IRay RC-09110X Datasheet	yes	−20°C to 150°C ±2°C/2%	0°C to 550°C ±3°C/3%	DVP, BT656, USB, CVBS	I2C	3.3V	<550mW, fast shot: <1.23W	-	48.7°	640×512	50	488€

Table 1: Comparison of different cameras

1.4.2 Finding a Micro-Controller and Powering it

After picking a thermal camera, it is necessary to pick a micro-controller that has the required peripherals and processing power. At this stage it is early to decide what micro-controller to use, as it is essential to prototype first with a computer.
Most micro-controllers need a 5V power source, so it is crucial to have a step-down converter with a output of 5V/3.3V to power the micro-controller and thermal camera.

1.4.3 Programming the Micro-Controller

The objective is to have a application where one could define a preset path for the robot to follow and set locations to observe with the thermal camera, but the application should at its most basic level be able to control the robot and gimbal with the thermal camera and see its video stream.

1.4.4 Assembly of all Components

From a mechanical standpoint, lightweight and rigid mounting structures must be designed to support the thermal camera, gimbal, micro-controller, and power electronics without compromising the robot's balance or mobility and minimize vibrations
Electrically, proper grounding, voltage regulation, and cable management are essential to prevent instability, or disconnections during movement.

1.5 Timeline

The following Gantt chart (**Figure 5**) outlines the main phases of the project, from initial research and component selection to system integration, testing, and final validation. Each task is structured to ensure a logical progression, with dependencies defined to maintain technical coherence and efficient resource allocation throughout the academic year.

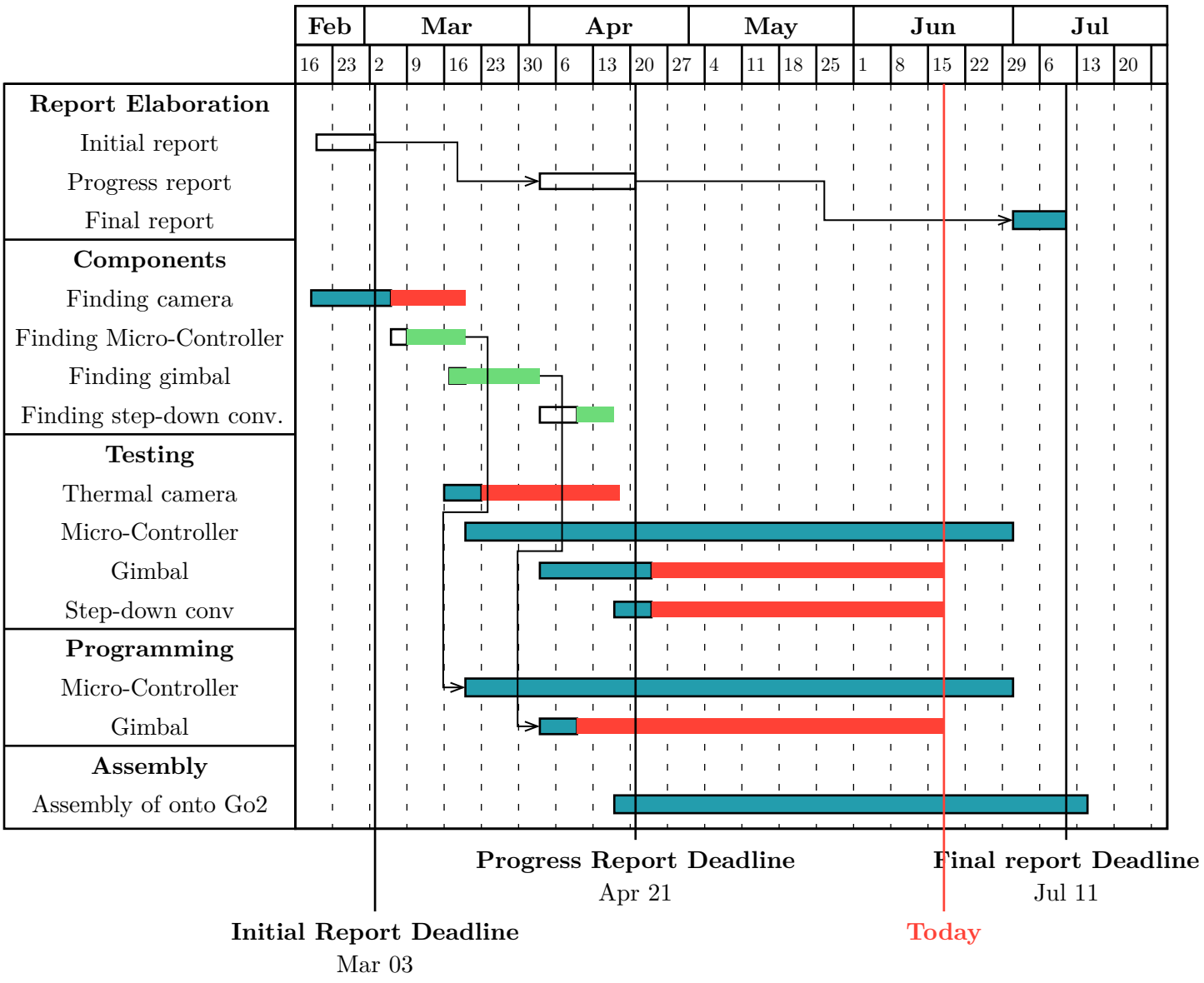


Figure 5: Gantt Chart

Chapter 2

System Architecture

This chapter details the selection and integration of the hardware components that constitute the robotic inspection system. It covers the rationale behind the choice of thermal camera, power electronics, and mechanical structures, as well as the assembly strategy for mounting all components onto the Unitree Go2 quadruped platform. Each subsystem is evaluated in terms of its technical specifications, compatibility with the overall architecture, and suitability for deployment in an industrial inspection context.

2.1 Component Selection

2.1.1 Thermal camera

Looking at **Table 1**, despite the higher cost, *IRay RC-09110X* seems the most attractive option, due to its high refresh rate and resolution, with the latter being essential for measuring temperatures of smaller objects like connections. But, as the camera comes from *Aliexpress*, the documentation is scarce and there is no SDK, so, the second best option is the *Lepton 3.5* which was selected due to its lower cost, availability, reduced FOV and radiometric capability, which enables absolute temperature measurements, which are essential for detecting overheating components in industrial environments. On contrary, the other higher-end modules provide improved spatial resolution, but their cost along with lack of documentation and very high FOV's make them undesirable.

This choice comes at the expense of a limited resolution of 160×120 pixels and a frame rate of 9 Hz, but it still remains adequate for the detection of temperature gradients and localized hotspots. The reduced performance is offset by its low power consumption, compact size, and cost-effectiveness, making it suitable for deployment on a mobile robotic platform where payload and energy constraints are relevant.

The *Lepton 3.5* module integrates into the Purethermal-3 development board which provides a *USB* interface that abstracts the native *SPI* and *I2C* (*CCI*) communication with the sensor. This significantly simplifies system development, allowing direct interaction with the camera using standard UVC (USB Video Class[8]) compatible tools and software libraries.



Figure 6: Image captured from the *Purethermal-3* board with the *Lepton 3.5* module

2.1.2 Step-down Converter

Given that the XT30 port on the *Unitree Go2* supplies a voltage that varies between 24V and 33.6V depending on battery state of charge, a robust and flexible DC-DC step-down converter is required to provide stable power to the onboard electronics.

The selected converter[9] is rated for 150W continuous output, accepting an input voltage range of 5V to 40V and providing an adjustable output between 1.2V and 35V. This range comfortably accommodates the full voltage swing of the battery while allowing the output to be set to a stable 5V rail, sufficient to power a single-board computer such as the *Raspberry Pi* (which draws up to 3A under load), *PureThermal-3*, *Lepton 3.5*, and the gimbal motors simultaneously.

The converter's wide input tolerance provides an additional safety margin against voltage spikes that may occur during dynamic motor operation of the robot. Its high power rating also ensures the system is not thermally stressed under peak current demand, improving long-term reliability in continuous inspection deployments.

2.1.3 Gimbal

As the design and manufacturing of a gimbal from scratch falls outside the scope of this project, an existing open-source model was sourced from Steve Eckerlein at *Printables*[10], a community platform for sharing 3D-printable designs. The selected model (**Figure 7**) provided a suitable mechanical baseline, a two-axis gimbal for a FPV camera that uses two SG90 servo motors, reducing development time and allowing focus on system integration rather than mechanical design.

As the original model was not dimensioned to accommodate the *PureThermal-3* development board and its attached *FLIR Lepton 3.5* module, modifications need to be made to the camera mount in order to match the mounting hole pattern of the *PureThermal-3* board.

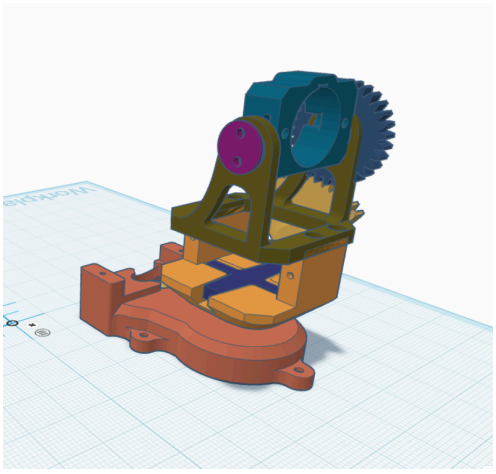


Figure 7: 3D assembly of the gimbal

2.2 Assembly onto the robot

To assemble the gimbal, step-down converter, micro-controller, and associated wiring onto the quadruped, it is essential to have a structural enclosure that can be mounted onto the *Unitree Go2* without permanent modifications to the robot's chassis. So, a 3D print cover plate design has been sourced from Gizmo Karl at *MakerWorld*[11] (**Figure 8** and **Figure 9**), specifically designed to fit the top surface of the *Unitree Go2* body. The part attaches via clip on and locks into the rail on the rear, requiring no permanent modifications to the robot's outer hull and allowing easy removal for maintenance or reconfiguration.
The use of 3D printing for the enclosure offers several practical advantages in the context of this project. Since the cover is fabricated from a digital model, it can be modified to accommodate the also 3D printed gimbal, the step-down converter, microcontroller and other eletronics.

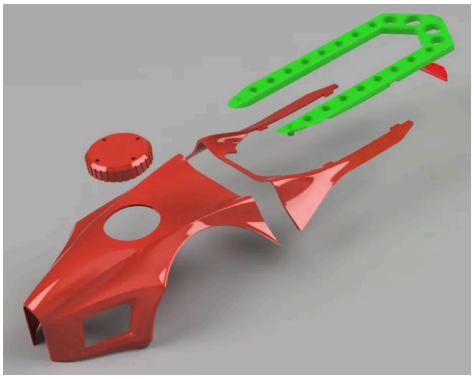


Figure 8: 3D render of the cover plate and rails



Figure 9: assembly of the cover plate and rails (3D print)

2.3 Programming

The primary programming objective of this project is to develop a interface that consolidates all monitoring and control functions into a single application. The final form of the application, whether a web-based or native desktop implementation, will provide the operator with three simultaneous video feeds: the thermal imaging feed from the *Lepton 3.5* along with the standard camera feed and the LiDAR point cloud visualization from the *Go2*'s onboard sensors.
In terms of control, the application will support robot movement via keyboard and mouse as the primary input method, with gamepad support as a desirable extension. Gimbal control will also be integrated, allowing the operator to aim the thermal camera independently of the robot's heading without needing to reposition the platform.

2.3.1 Robot Control

Because of the mobile App, the *Unitree Go2* exposes a *WebRTC*-based communication interface that allows external applications to interact with the robot over a wireless network. The specific library used in this project is *unitree_webrtc_connect* [12], an open-source Python package that reverse-engineered the underlying *WebRTC* protocol, allowing more freedom over the official *Unitree SDK* because interfaces like

sport mode (what allows the robot to move and perform actions like sitting, standing, etc…) are locked to the EDU model.

2.3.2 Thermal Camera Interface

The *PureThermal-3* development board exposes the *Lepton 3.5* to the host system as a standard *UVC* (USB Video Class [8]) device, meaning it is recognised by the operating system without the need for proprietary drivers. This allows the video stream to be captured using standard tools such as *Video for Linux* (*V4L2*) or *OpenCV*, simplifying integration into a Python-based application.

Beyond basic video capture, full access to the *Lepton 3.5*'s radiometric and configuration capabilities requires interaction with the *CCI* interface, which the *PureThermal-3* exposes through *UVC* extension units, allowing the host to read and write sensor registers without direct *CCI* access.On *Linux*, this can be achieved through the standard *V4L2* API using *uvcdynctrl* [13], or more portably through *libuvc* [14], a cross-platform C library for *USB* video devices.

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